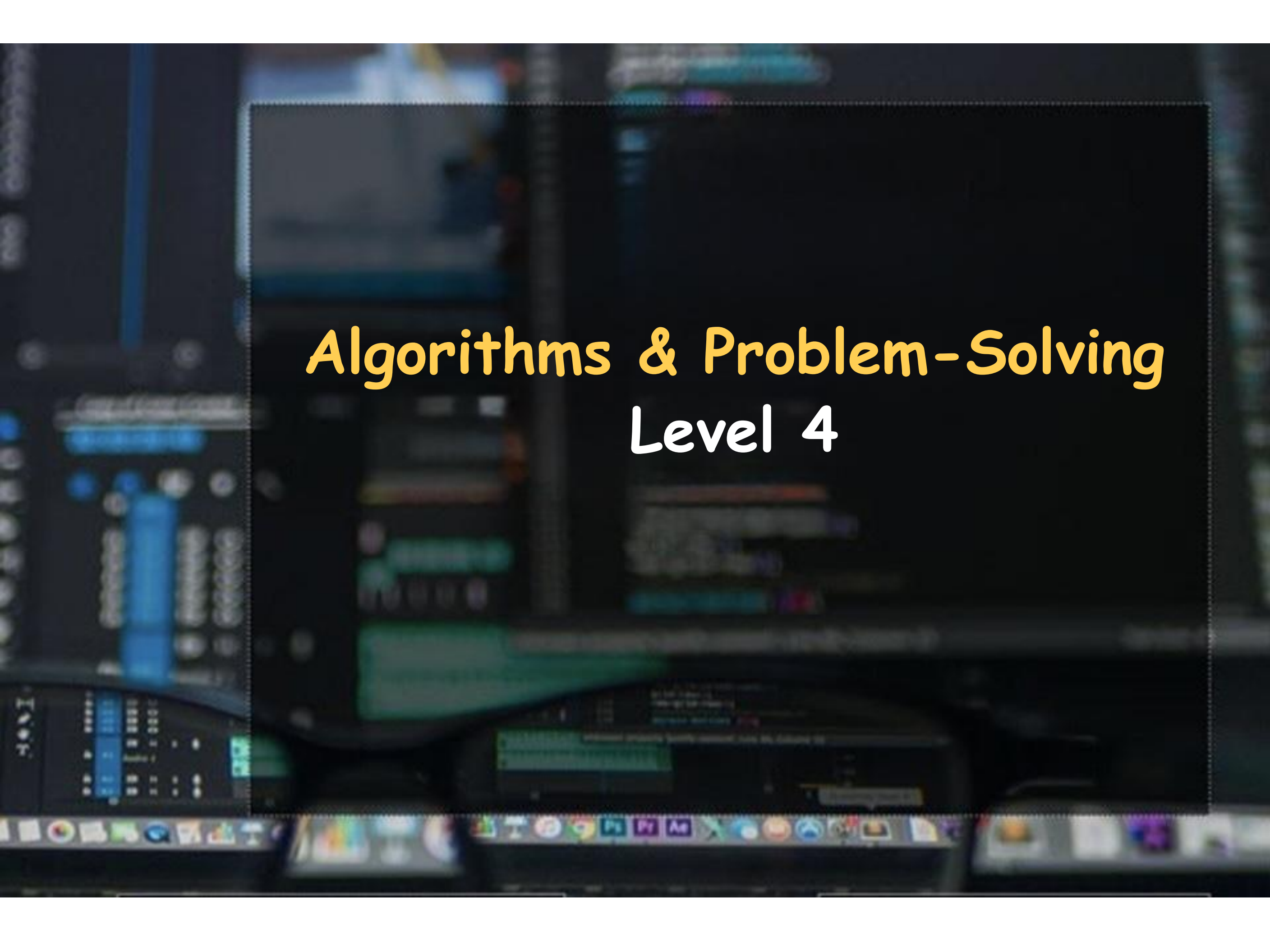
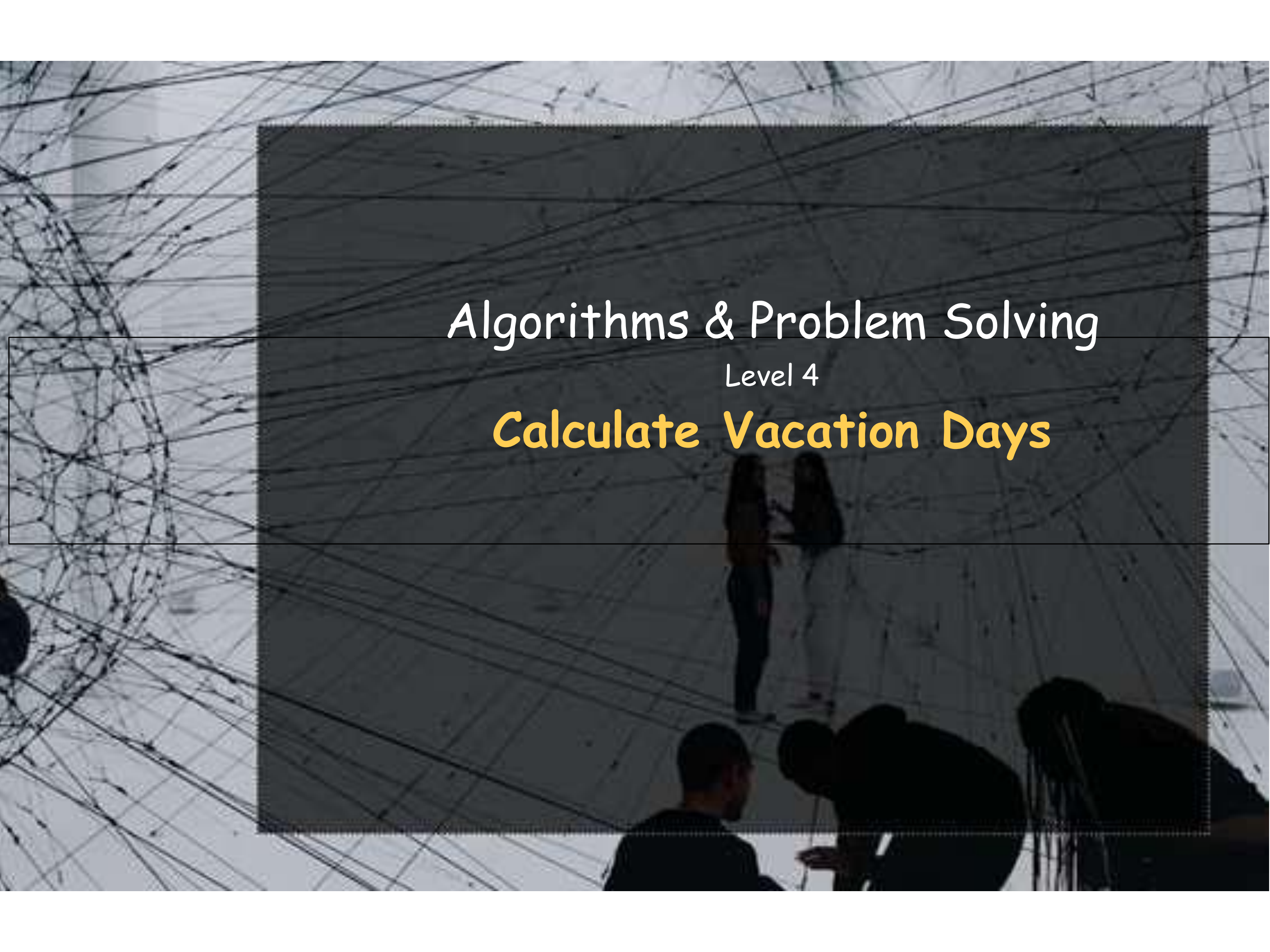




Algorithms & Problem-Solving Level 4

The background of the slide features a complex network diagram with numerous nodes and connecting lines, overlaid on a dark, semi-transparent rectangular area. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative work environment, possibly looking at a screen or a large diagram.

Algorithms & Problem Solving

Level 4

Calculate Vacation Days



Definition

- The goal is to create a C++ program that calculates the actual number of vacation days between two dates (from - to), excluding weekends (Friday and Saturday) from the calculation.

Explanation:

-  The program uses the `sDate` struct data structure to represent the date, and includes several helper functions to perform time operations.

Procedures

- 1 Read the dates from the user
 -  ReadFullDate() Reads the day, month, and year and returns an sDate object for both the start and end dates.
- 2 Check for leap years
 -  IsLeapYear(year)
 -  Used to calculate the number of days in February (29 instead of 28 in a leap year).
- 3 Determine the number of days in a month
 -  NumberOfDaysInAMonth(month, year)
 -  Used to determine the end of the month in subsequent functions.
- 4 Increment a date by one day
 -  IncreaseDateByOneDay(date)
 -  Used to increment a day during a holiday.
- 5 Determine the day of the week
 -  DayOfWeekOrder(day, month, year)
 -  Returns the day order (0 for Sunday, 6 for Saturday).
 -  DayShortName() Returns Abbreviated day name (e.g., Sun, Mon...)
- 6 Day type check
 -  IsWeekEnd(date) checks if it's Friday or Saturday
 -  IsBusinessDay(date) checks if it's a workday (Sunday to Thursday)

- 7 Actual vacation days calculation
 - CalculateVacationDays(dateFrom, dateTo)
 - Starts from dateFrom to dateTo
 - For each day, the following is checked:
 - If it's a workday, it's added to the counter
 - Then it moves to the next day



Example:

Vacation Starts:

Please enter a Day? 1

Please enter a Month? 9

Please enter a Year? 2022

Vacation Ends:

Please enter a Day? 5

Please enter a Month? 9

Please enter a Year? 2022

Vaction From: Thu, 1/9/2022

Vaction To: Mon, 5/9/2022

Actual Vacation Days is: 2



Note:

- The count excludes weekends.
- The program calculates the days between the two dates, excluding the last day if it is not a workday.

Explanation:

- Thursday (1/9) = Working day
- Friday and Saturday are holidays
- Sunday and Monday are working days
- Total = 2 working days

Summary

- ✓ The program applies the following concepts:
 - Working with date structures
 - Checking the day type (work/holiday)
 - Using loops and helper functions to calculate time differences
 - Practical application of conditional logic skills in programming

Thank you 😊